Anomaly Detection Report

1. Introduction:

The purpose of anomaly detection was to identify unusual boost pressure readings in the locomotive dataset.

The analysis focused on two important features:

- speed
- boost_pressure

These features were selected to examine the relationship between locomotive speed and boost pressure and to identify unusual pressure readings.

2. Dataset Preparation:

The dataset was loaded from a ZIP file containing multiple CSV files.

The following steps were performed:

- Loaded and combined 20 CSV files.
- Created one master dataset.
- Selected timestamp, locomotive, speed, and boost_pressure.
- Removed rows containing missing values in the selected columns.

Dataset Summary:

- Total rows before cleaning: 38,882,843
- Total rows after cleaning: 31,694,963
- Total columns in the original dataset: 41
- Locomotives analyzed: 5

The locomotive IDs were:

- 50904
- 54904
- 55904
- 56904
- 59904

3. Method Used:

For anomaly detection, we used **Linear Regression and Residual-Based Detection**.

Linear regression estimates the expected boost pressure based on engine speed.

The formula is:

Predicted Pressure = m × Speed + c

Where:

- m = slope
- c = intercept

Then we calculate the absolute difference between actual and predicted pressure:

Residual = |Actual Pressure − Predicted Pressure|

The anomaly threshold is:

Threshold = 3 × Standard Deviation of Residuals

Any reading is marked as a potential anomaly if:

- Boost pressure is equal to zero, or
- The residual is greater than the threshold.

5. Linear Regression:

Initially, linear regression was applied to the complete cleaned dataset.

The regression equation was:

Pressure = 0.5544 × Speed + 4.4259

The threshold was calculated as:

99.89

The initial model detected:

2,493,531 potential anomalies (7.87% of the cleaned data).

This model provides an overall estimate of the relationship between speed and boost pressure.

6. Locomotive-Wise Anomaly Detection:

After the initial analysis, anomaly detection was performed separately for each locomotive.

For each locomotive:

- A separate linear regression model was created.
- Predicted boost pressure was calculated.
- Residuals were calculated.
- A separate anomaly threshold was determined.
- Zero-pressure readings and readings above the threshold were marked as anomalies.

Results:

ANOMALY & NORMAL SAMPLE TABLE

Speed	Locomotive	Normal Pressure	Boost Pressure	Status
10.0	50904	17.26	0.86	Normal
10.0	59904	5.76	0.00	Anomaly
30.0	50904	27.83	16.38	Normal
30.0	50904	27.83	180.16	Anomaly
50.0	50904	37.55	1.72	Normal
50.0	54904	31.73	193.09	Anomaly
70.0	59904	38.01	114.65	Normal
70.0	50904	48.79	183.61	Anomaly
90.0	54904	54.47	64.65	Normal
90.0	50904	58.12	191.36	Anomaly

Timestamp-Based Comparison

[Anomaly Found] Locomotive 50904 at Speed 30.0 | Timestamp: 2025-12-06 11:27:13+00:00
Checking what OTHER engines were exhibiting at this exact second:

timestamp	locomotive	speed	boost_pressure
2025-12-06 11:27:13+00:00	50904	30.1860	180.158
2025-12-06 11:27:13+00:00	54904	30.5568	179.296
2025-12-06 11:27:13+00:00	55904	30.9276	71.546
2025-12-06 11:27:13+00:00	56904	32.2236	179.296
2025-12-06 11:27:13+00:00	59904	30.9276	62.926

```
[Anomaly Found] Locomotive 54904 at Speed 50.0 | Timestamp: 2025-12-11 20:28:19+00:00
 Checking what OTHER engines were exhibiting at this exact second:
                 timestamp  locomotive   speed  boost_pressure
2025-12-11 20:28:19+00:00       50904 50.3748         184.468
2025-12-11 20:28:19+00:00       54904 49.2624         193.088
2025-12-11 20:28:19+00:00       55904 51.8544         169.814
2025-12-11 20:28:19+00:00       56904 51.3000         193.950
2025-12-11 20:28:19+00:00       59904 49.6332         174.986
```

```
[Anomaly Found] Locomotive 50904 at Speed 70.0 | Timestamp: 2025-11-15 18:13:47+00:00
 Checking what OTHER engines were exhibiting at this exact second:
                 timestamp  locomotive   speed  boost_pressure
2025-11-15 18:13:47+00:00       50904 70.9308         183.606
2025-11-15 18:13:47+00:00       54904 70.1892         187.054
2025-11-15 18:13:47+00:00       55904 71.1180          55.168
2025-11-15 18:13:47+00:00       56904 71.1180         190.502
2025-11-15 18:13:47+00:00       59904 70.9308         101.716
```

```
[Anomaly Found] Locomotive 59904 at Speed 10.0 | Timestamp: 2025-12-10 06:27:16+00:00
 Checking what OTHER engines were exhibiting at this exact second:
                 timestamp  locomotive  speed  boost_pressure
2025-12-10 06:27:16+00:00       50904 9.6300           1.724
2025-12-10 06:27:16+00:00       54904 9.4464           3.448
2025-12-10 06:27:16+00:00       55904 9.2592           0.862
2025-12-10 06:27:16+00:00       56904 9.4464           5.172
2025-12-10 06:27:16+00:00       59904 9.8172           0.000
```

```
[Anomaly Found] Locomotive 50904 at Speed 90.0 | Timestamp: 2025-10-23 21:34:08+00:00
 Checking what OTHER engines were exhibiting at this exact second:
                 timestamp  locomotive   speed  boost_pressure
2025-10-23 21:34:08+00:00       50904 89.0820         191.364
2025-10-23 21:34:08+00:00       54904 89.8236          72.408
2025-10-23 21:34:08+00:00       55904 89.6364         187.054
2025-10-23 21:34:08+00:00       56904     NaN             NaN
2025-10-23 21:34:08+00:00       56904 89.6364         193.088
2025-10-23 21:34:08+00:00       59904 88.8948         174.986
```